

GROUP 6 PRESENTATION

COURSE CODE: INS 202

COURSE TITLE: HUMAN COMPUTER INTERFACE

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ASSIGNMENT 1

Questions:

- Identify a system
- Identify the real and perceived affordances
- Identify the goal and constraints of the system

ANSWERS

1. Identified System: An Automated Vending Machine

An Automated Vending Machine is a self-service system that integrates multiple components a physical cabinet, a mechanical dispensing unit, an electronic payment processor, and a user interface — to allow customers to purchase goods without any human attendant. These machines are found in schools, hospitals, airports, and office buildings, offering round-the-clock access to snacks, beverages, and other retail items.

The design of its interface must simultaneously accommodate first-time users with zero instruction, while being efficient enough for experienced users.

2a. Real Affordances (Physical / Foundational Capabilities)

Real affordances refer to the actual, physical properties of a system that determine how it can be used — regardless of whether the user is aware of them. They are rooted in the material reality of the machine.

- **Coin / Card Entry Slot:** The machine is equipped with dedicated physical openings or contactless sensor arrays that are engineered specifically to receive, authenticate, and process currency (coins and notes) or electronic payment signals (NFC, debit/credit cards).
- **Refrigeration Unit :** Some vending machines house compressor-based cooling systems that maintain temperature-sensitive products (beverages, dairy) at safe temperatures. This is a real affordance that enables the machine to stock perishable goods.
- **Product Dispensing:** Mechanical coils, belts, robotic arms, physically move items from storage to a delivery bin
- **Selection input:** Buttons or touchscreens allow the user to send a specific electronic signal representing a product code

2b. Perceived Affordances (How the User Thinks It Works)

Perceived affordances : Also called 'perceived' or 'psychological' affordances are the cues and signals a design communicates to users about how to interact with it. They are the mental model the interface creates in the user's mind, regardless of the underlying mechanics.

- **Transparent Glass Panel:** The large glass front of the machine is not merely structural; it functions as a visual catalog. By making the inventory directly visible, the glass communicates 'these items are available for selection.'
- **Delivery Flap at the Base:** The hinged, spring-loaded flap at the bottom of the machine creates a strong affordance. Its location, size, and tactile resistance communicate: 'reach in here to retrieve your item.' Users instinctively move to the bottom of the machine after making a selection. The opening itself signals completion of the transaction.
- **Backlit Keypad / Touch Display:** Illuminated numbers, letters, or on-screen icons direct the user's visual attention and signal interactivity. The backlighting communicates 'this is where you interact.' The layout of the keypad — alphanumeric.
- **Prominent 'Cancel' or 'Return' Button (often red):** The color coding and position of this button affords the mental model that a transaction can be reversed.

3. Goal of the System

The primary goal of the Automated Vending Machine is to provide continuous (24/7), autonomous, and efficient access to retail goods while minimizing operational costs through the elimination of human attendants. The system bridges the gap between traditional retail environments and the modern consumer's need for instant, low-friction purchasing.

The machine must make the transaction feel intuitive and error-free across a diverse user population — including children, the elderly, non-native speakers, and first-time users. This demands that the interface be self-explanatory, fault-tolerant, and efficient.

4. Constraints of the System

Constraints in HCI and systems design are the factors that limit the degrees of freedom of both the system and its users. Vending machines operate under four primary categories of constraint:

Physical Constraints:

- The internal dispensing mechanism (coil diameter, shelf dimensions) restricts the physical size and shape of products that can be stocked. A bag of chips must fit between the mechanical coils without jamming; a bottle must not shatter when dispensed.
- The machine's footprint determines installation locations — only spaces with sufficient floor space, wall clearance, and ventilation qualify.

Technical Constraints:

- The machine requires a stable electrical power supply for operation, refrigeration, and screen illumination. Power outages render the system non-functional.
- Card payment and remote monitoring features require an active internet or cellular network connection. In areas with poor connectivity, these features fail, reverting the machine to cash-only mode or offline operation.

Security and Maintenance Constraints:

- The machine is susceptible to physical vandalism, theft (tilting attacks), and electronic fraud (skimming devices on card readers). These risks constrain deployment to supervised locations or require additional security hardware.
- Mechanical jams — where a product becomes lodged mid-dispensing — are a common failure mode. These require manual intervention by a technician, creating downtime and user frustration.

Inventory Constraints:

- Once a product row is depleted, the machine cannot fulfil further requests for that item until it is physically restocked. This creates a mismatch between user demand and system capability, a constraint the system communicates through 'Sold Out' indicators on the selection display.

ASSIGNMENT 2

Questions:

- Identify the system
- Identify the users
- Identify the goal
- Identify benchmarks
- Identify the phases of design

1. Identified System: A Smart Learning Thermostat

A Smart Learning Thermostat is an Internet of Things (IoT) system designed to automate home temperature management. It consists of three integrated components: a wall-mounted physical interface (the device itself), a companion mobile application (iOS/Android), and a distributed network of environmental sensors.

Unlike traditional thermostats — which require manual adjustment every time comfort conditions change — the smart thermostat uses machine learning algorithms to build a predictive model of household routines. It learns when occupants wake up, leave for work, return home, and go to sleep, automatically adjusting temperature to match these patterns without requiring ongoing user input.

It also integrates with external systems such as Google Home, Amazon Alexa, and local weather APIs, enabling it to proactively respond to forecast changes. For example, if outdoor temperatures are expected to drop sharply overnight, the thermostat can pre-heat the home before the occupants feel the cold.

2. The Users

The smart thermostat serves a heterogeneous user population with significantly different technical abilities and interaction patterns. HCI design must address all of them simultaneously.

Primary User — Tech-Savvy Head of Household:

- This user manages energy bills and takes responsibility for configuring the system. They are comfortable navigating app-based settings, creating custom schedules, interpreting energy usage reports, and integrating the thermostat with other smart home devices.
- Their interaction is high-frequency and high-complexity — they engage deeply with the system during initial setup and periodically review performance analytics.

Secondary Users — Other Household Members:

- Children and elderly residents form the secondary user group. Their interactions are simple and infrequent — a manual temperature adjustment when feeling too hot or cold.
- For this group, the interface must be immediately understandable without any configuration knowledge. Large, clearly labelled controls and minimal steps to accomplish a temperature change are critical HCI requirements.

3. Goal of the System

The overarching goal of the Smart Learning Thermostat is to maintain optimal home comfort while simultaneously maximising energy efficiency. These two goals exist in tension — comfort often requires more energy — and the thermostat's intelligence lies in resolving this tension in a user-specific way.

Specific Goal Dimensions:

- **Cognitive Load Reduction:** The system's learning capability is fundamentally an HCI feature. By 'remembering' that the user leaves for work at 8:00 AM and returns at 6:00 PM, the thermostat eliminates the recurring cognitive task of adjusting the temperature.
- **Energy Efficiency:** The system aims to reduce monthly heating and cooling costs by avoiding unnecessary conditioning of empty spaces. This serves both the household's financial interests and the broader societal goal of reducing carbon emissions.
- **Proactive Comfort Management:** Rather than reacting to user discomfort, the system anticipates needs. Pre-cooling a home before occupants arrive in summer — based on learned patterns and weather.

4. Benchmarks

Benchmarks are quantitative standards used to evaluate whether a system meets its design goals, and to compare performance across versions or competitors. For the Smart Learning Thermostat, four categories of benchmark are relevant:

Efficiency Benchmarks:

- A 10-15% reduction in monthly heating and cooling costs compared to a conventional manual thermostat. This is the primary value proposition of the product.
- After 7 days of manual use, the system should demonstrate measurable predictive accuracy — automatically adjusting temperatures within 30 minutes of the user's usual routine.

Usability Benchmarks:

- A first-time user should be able to set a weekly schedule in under 2 minutes without consulting a user manual. This tests the interface's learnability.

- New users should complete initial setup (Wi-Fi connection, room configuration, account creation) in under 10 minutes.

Reliability Benchmarks:

- The system should maintain 99.5% uptime for local control (manual adjustment on the wall unit) even when internet connectivity is lost.
- Sensor accuracy should remain within $\pm 0.5^{\circ}\text{C}$ across the full operational temperature range.

5. Phases of Design

The design of the Smart Learning Thermostat follows an iterative, user-centred design process. Each phase builds on the previous, incorporating user feedback to refine both the hardware and software interface.

Phase 1 — User Research and Requirements Gathering:

Designers conduct ethnographic research in real homes to understand how different household members interact with temperature controls. Interviews, diary studies, and observation reveal pain points with existing systems — for example, the frustration of returning to a cold house, or the difficulty elderly users have with digital interfaces. Functional requirements (what the system must do) and non-functional requirements (response time, accuracy) are formally documented.

Phase 2 — Conceptual Design and Prototyping:

Design teams create low-fidelity prototypes of both the physical device and the mobile app. Paper mockups and wireframes are tested with representative users to evaluate conceptual clarity. Key questions at this phase include: Do users understand what the 'leaf' (energy-saving) icon means? Can elderly users find the manual override without assistance? Feedback drives the selection of interface metaphors and layout patterns.

Phase 3 — Detailed Design and Implementation:

High-fidelity prototypes and working software are developed. The machine learning algorithm is trained on anonymised household data sets. Hardware components — sensors, display, physical ring control — are finalized and tested for manufacturing tolerances. Accessibility features (large text option, high-contrast mode, voice control integration) are implemented to accommodate secondary users.

Phase 4 — Iterative Testing and Evaluation:

. Beta units are deployed in real homes, and designers collect both quantitative data (energy savings, error rates, session duration) and qualitative feedback (user satisfaction surveys, interviews). Pain points are identified and categorized by severity.

- A/B Testing: Half the beta testers receive a unit with a touchscreen interface; the other half receive a unit with a physical rotary dial. Error rates and user preference scores are compared to determine which modality better serves the target population.

- Pain Point Discovery: If users are repeatedly overriding the auto-schedule, this signals that the learning algorithm is not accurately modelling their preferences. The team uses this data to retrain the model with a more granular input dataset.
- Icon Comprehension Testing: If a significant proportion of users misinterpret the 'leaf' icon as a 'nature mode' rather than 'energy saving mode,' designers add a contextual tooltip or replace the icon entirely in the next iteration.

SUMMARY:

Vending machine was chosen because of Universal Accessibility :

It is intuitive enough for anyone to use, first time users, children and elders with zero prior instruction. Fills the gap between the traditional retail and modern need for 24/7b autonomous access, transaction that are made seamlessly .

WHILE, Smart learning thermostat was picked because it is an Internet of things [IoT] , and it shows the modern shift towards intelligent automated interfaces, show proactive interaction where the system learns and anticipates the users needs.

A prefect example of a system that eliminates recurring tasks through machine learning REMEMBERING user's routine to automate comfort

Thank You.